

Exponential Functions & Growth Models

ANSWER KEY



Exponential growth and decay basics

- 1 Classify each function as exponential growth or decay, and state the growth/decay factor: $f(x) = 5(1.2)^x$ and $g(x) = 5(0.8)^x$.

$f(x) = 5(1.2)^x$: growth, since the base $1.2 > 1$ (growth rate 20% per unit x). $g(x) = 5(0.8)^x$: decay, since the base $0.8 < 1$ (decay rate 20% per unit x).

- 2 A population of bacteria is modeled by $P(t) = 1000(2)^{t/3}$, where t is in hours. Interpret the meaning of the exponent's structure, and find the population after 9 hours.

The form $2^{t/3}$ means the population doubles every 3 hours. After 9 hours (3 doubling periods):
 $P(9) = 1000(2)^3 = 1000(8) = 8000$ bacteria.

Visualizing exponential growth

- 3 The graph shows $f(x) = 2^x$ and $g(x) = 3(0.5)^x$, one growing and one decaying.

a Identify which curve represents growth and which represents decay. $f(x) = 2^x$ is the growth curve (increasing, base > 1). $g(x) = 3(0.5)^x$ is the decay curve (decreasing, base < 1).

b State the horizontal asymptote shared by both curves. Both curves approach the horizontal asymptote $y = 0$ (neither ever reaches or crosses it).

Finding an exponential model from data

- 4 An investment grows from 2000 dollars to 2420 dollars in 2 years under exponential growth $A(t) = A_0(1 + r)^t$. Find the annual growth rate r .

$2420 = 2000(1 + r)^2$: $(1 + r)^2 = 1.21$, so $1 + r = \sqrt{1.21} = 1.1$, giving $r = 0.10 = 10\%$.

- 5 A radioactive sample decays from 80 g to 45 g over 6 years. Find the annual decay factor b in the model $A(t) = 80b^t$, to four decimal places.

$45 = 80b^6$: $b^6 = \frac{45}{80} = 0.5625$. $b = (0.5625)^{1/6} \approx 0.9128$.

Continuous growth and the number e

- 6 A quantity grows continuously according to $A(t) = 500e^{0.03t}$. Find the amount after 20 years, to the nearest whole number.

$$A(20) = 500e^{0.03(20)} = 500e^{0.6} \approx 500(1.8221) \approx 911.$$

- 7 Explain the difference between the discrete growth model $A(t) = A_0(1 + r)^t$ and the continuous growth model $A(t) = A_0e^{kt}$, and state how r and k are related when both models describe the same annual growth rate r .

The discrete model compounds growth once per period (e.g. once a year), while the continuous model compounds growth constantly at every instant. They're related by $e^k = 1 + r$, i.e. $k = \ln(1 + r)$ — the continuous rate k is slightly less than the discrete rate r for positive growth, since continuous compounding reaches the same total growth with a slightly lower nominal rate.

Comparing growth rates: exponential vs. polynomial

- 8 Explain why, for large enough x , any exponential growth function $f(x) = b^x$ (with $b > 1$) will eventually exceed any polynomial function $g(x) = x^n$, no matter how large the exponent n is. Illustrate with $f(x) = 1.01^x$ vs. $g(x) = x^{10}$.

Exponential functions grow by repeated multiplication, so their rate of growth itself keeps growing exponentially, while a polynomial's growth rate eventually stabilizes into a fixed power-law pattern. Even a very slowly-growing exponential like 1.01^x will eventually overtake x^{10} , because exponential growth compounds indefinitely while polynomial growth does not — it just takes an enormous value of x (in this case, extremely large, since 1.01^x grows very slowly at first) for the crossover to occur.

Applications: half-life and doubling time

- 9 A substance has a half-life of 8 days. Write its decay model $A(t) = A_0\left(\frac{1}{2}\right)^{t/8}$, then find what fraction of the original amount remains after 20 days.

$$\frac{A(20)}{A_0} = \left(\frac{1}{2}\right)^{20/8} = \left(\frac{1}{2}\right)^{2.5} = 2^{-2.5} \approx 0.1768, \text{ so about } 17.68\% \text{ remains.}$$

- 10 A city's population doubles every 15 years and currently stands at 40,000. Write an exponential model $P(t)$ and predict the population after 45 years.

$$P(t) = 40000(2)^{t/15}. \text{ After 45 years } (t/15 = 3 \text{ doubling periods}): \\ P(45) = 40000(2)^3 = 40000(8) = 320,000.$$

Transformations of exponential functions

- 11 Describe how the graph of $g(x) = 2^{x-3} + 4$ is transformed from the parent function $f(x) = 2^x$, and state the new horizontal asymptote.

$g(x) = 2^{x-3} + 4$ shifts $f(x) = 2^x$ right 3 units and up 4 units. The horizontal asymptote shifts from $y = 0$ to $y = 4$.

- 12 Find the horizontal asymptote and y -intercept of $h(x) = -3(2)^x + 6$.

The negative coefficient reflects the graph over the horizontal asymptote line; as $x \rightarrow \infty$, $-3(2)^x \rightarrow -\infty$, so $h(x) \rightarrow -\infty$, while as $x \rightarrow -\infty$, $-3(2)^x \rightarrow 0$, giving horizontal asymptote $y = 6$. y -intercept: $h(0) = -3(1) + 6 = 3$, so $(0, 3)$.

Compound interest and financial models

- 13 A savings account earns 4% annual interest compounded quarterly, modeled by $A(t) = A_0 \left(1 + \frac{0.04}{4}\right)^{4t}$. If 5000 dollars is deposited, find the balance after 3 years, to the nearest dollar.

$$A(3) = 5000(1.01)^{12} \approx 5000(1.1268) \approx 5634 \text{ dollars.}$$

- 14 Explain, without computing exact values, why compounding more frequently (e.g. daily instead of quarterly) at the same nominal annual rate produces a slightly larger balance after the same number of years.

More frequent compounding applies the (smaller) periodic interest rate more often, and interest starts earning interest sooner and more often within each year. This pushes the effective annual growth rate slightly above the nominal rate, so daily compounding yields a marginally larger balance than quarterly compounding at the same stated annual rate.

Multiple choice — easy 15

- 15 A population is modeled by $P(t) = 900(3)^{t/3}$, where t is in years. Find $P(6)$.

- a. 2700
- b. 5400
- c. 8100 ✓
- d. 24300

Since $t/3 = 2$: $P(6) = 900(3)^2 = 900(9) = 8100$.

16 A population is modeled by $P(t) = 200(3)^{t/6}$, where t is in years. Find $P(6)$.

a. 800

b. 600 ✓

c. 400

d. 1800

Since $t/6 = 1$: $P(6) = 200(3)^1 = 200(3) = 600$.

17 A population is modeled by $P(t) = 700(3)^{t/3}$, where t is in years. Find $P(3)$.

a. 2800

b. 6300

c. 1400

d. 2100 ✓

Since $t/3 = 1$: $P(3) = 700(3)^1 = 700(3) = 2100$.

18 A population is modeled by $P(t) = 400(2)^{t/7}$, where t is in years. Find $P(7)$.

a. 1200

b. 1600

c. 800 ✓

d. 400

Since $t/7 = 1$: $P(7) = 400(2)^1 = 400(2) = 800$.

19 A population is modeled by $P(t) = 700(2)^{t/2}$, where t is in years. Find $P(4)$.

a. 5600

b. 1400

c. 2100

d. 2800 ✓

Since $t/2 = 2$: $P(4) = 700(2)^2 = 700(4) = 2800$.

20 A population is modeled by $P(t) = 700(2)^{t/2}$, where t is in years. Find $P(6)$.

a. 11200

b. 4200

c. 2800

d. 5600 ✓

Since $t/2 = 3$: $P(6) = 700(2)^3 = 700(8) = 5600$.

21 A population is modeled by $P(t) = 500(3)^{t/3}$, where t is in years. Find $P(3)$.

a. 1000

b. 2500

c. 4500

d. 1500 ✓

Since $t/3 = 1$: $P(3) = 500(3)^1 = 500(3) = 1500$.

22 A population is modeled by $P(t) = 500(2)^{t/4}$, where t is in years. Find $P(16)$.

a. 9000

b. 8000 ✓

c. 16000

d. 4000

Since $t/4 = 4$: $P(16) = 500(2)^4 = 500(16) = 8000$.

23 A population is modeled by $P(t) = 100(2)^{t/4}$, where t is in years. Find $P(8)$.

a. 800

b. 400 ✓

c. 200

d. 300

Since $t/4 = 2$: $P(8) = 100(2)^2 = 100(4) = 400$.

24 A population is modeled by $P(t) = 900(3)^{t/5}$, where t is in years. Find $P(15)$.

a. 26100

b. 8100

c. 24300 ✓

d. 72900

Since $t/5 = 3$: $P(15) = 900(3)^3 = 900(27) = 24300$.

25 A population is modeled by $P(t) = 700(3)^{t/2}$, where t is in years. Find $P(6)$.

a. 20300

b. 56700

c. 18900 ✓

d. 6300

Since $t/2 = 3$: $P(6) = 700(3)^3 = 700(27) = 18900$.

26 A population is modeled by $P(t) = 200(2)^{t/5}$, where t is in years. Find $P(20)$.

a. 3200 ✓

b. 6400

c. 1600

d. 3600

Since $t/5 = 4$: $P(20) = 200(2)^4 = 200(16) = 3200$.

27 A population is modeled by $P(t) = 800(3)^{t/2}$, where t is in years. Find $P(2)$.

a. 4000

b. 2400 ✓

c. 3200

d. 7200

Since $t/2 = 1$: $P(2) = 800(3)^1 = 800(3) = 2400$.

28 A population is modeled by $P(t) = 500(2)^{t/5}$, where t is in years. Find $P(20)$.

a. 8500

b. 8000 ✓

c. 4000

d. 16000

Since $t/5 = 4$: $P(20) = 500(2)^4 = 500(16) = 8000$.

29 A population is modeled by $P(t) = 600(2)^{t/6}$, where t is in years. Find $P(12)$.

a. 4800

b. 2400 ✓

c. 1200

d. 1800

Since $t/6 = 2$: $P(12) = 600(2)^2 = 600(4) = 2400$.

Multiple choice — medium 15

30 A substance has a half-life of 6 days. What fraction of the original amount remains after 18 days?

- a. $\frac{1}{4}$
- b. $\frac{1}{16}$
- c. $\frac{3}{8}$
- d. $\frac{1}{8}$ ✓

Since $t/6 = 3$ half-lives have passed: fraction remaining $= \left(\frac{1}{2}\right)^3 = \frac{1}{8}$.

31 A substance has a half-life of 6 days. What fraction of the original amount remains after 30 days?

- a. $\frac{1}{16}$
- b. $\frac{5}{32}$
- c. $\frac{1}{32}$ ✓
- d. $\frac{1}{64}$

Since $t/6 = 5$ half-lives have passed: fraction remaining $= \left(\frac{1}{2}\right)^5 = \frac{1}{32}$.

32 A substance has a half-life of 2 days. What fraction of the original amount remains after 2 days?

- a. $\frac{1}{4}$
- b. $\frac{5}{8}$
- c. $\frac{1}{2}$ ✓
- d. $\frac{3}{4}$

Since $t/2 = 1$ half-lives have passed: fraction remaining $= \left(\frac{1}{2}\right)^1 = \frac{1}{2}$.

33 A substance has a half-life of 5 days. What fraction of the original amount remains after 15 days?

- a. $\frac{1}{16}$
- b. $\frac{1}{4}$
- c. $\frac{1}{8}$ ✓
- d. $\frac{3}{8}$

Since $t/5 = 3$ half-lives have passed: fraction remaining $= \left(\frac{1}{2}\right)^3 = \frac{1}{8}$.

34 A substance has a half-life of 2 days. What fraction of the original amount remains after 6 days?

a. $\frac{1}{4}$

b. $\frac{1}{8}$ ✓

c. $\frac{1}{16}$

d. $\frac{3}{8}$

Since $t/2 = 3$ half-lives have passed: fraction remaining $= \left(\frac{1}{2}\right)^3 = \frac{1}{8}$.

35 A substance has a half-life of 6 days. What fraction of the original amount remains after 12 days?

a. $\frac{5}{16}$

b. $\frac{1}{8}$

c. $\frac{1}{4}$ ✓

d. $\frac{1}{2}$

Since $t/6 = 2$ half-lives have passed: fraction remaining $= \left(\frac{1}{2}\right)^2 = \frac{1}{4}$.

36 A substance has a half-life of 8 days. What fraction of the original amount remains after 24 days?

a. $\frac{1}{16}$

b. $\frac{1}{4}$

c. $\frac{3}{8}$

d. $\frac{1}{8}$ ✓

Since $t/8 = 3$ half-lives have passed: fraction remaining $= \left(\frac{1}{2}\right)^3 = \frac{1}{8}$.

37 A substance has a half-life of 7 days. What fraction of the original amount remains after 21 days?

a. $\frac{1}{8}$ ✓

b. $\frac{3}{8}$

c. $\frac{1}{16}$

d. $\frac{1}{4}$

Since $t/7 = 3$ half-lives have passed: fraction remaining $= \left(\frac{1}{2}\right)^3 = \frac{1}{8}$.

38 A substance has a half-life of 7 days. What fraction of the original amount remains after 35 days?

a. $\frac{1}{32}$ ✓

b. $\frac{5}{32}$

c. $\frac{1}{64}$

d. $\frac{1}{16}$

Since $t/7 = 5$ half-lives have passed: fraction remaining $= \left(\frac{1}{2}\right)^5 = \frac{1}{32}$.

39 A substance has a half-life of 3 days. What fraction of the original amount remains after 9 days?

a. $\frac{1}{8}$ ✓

b. $\frac{3}{8}$

c. $\frac{1}{4}$

d. $\frac{1}{16}$

Since $t/3 = 3$ half-lives have passed: fraction remaining $= \left(\frac{1}{2}\right)^3 = \frac{1}{8}$.

40 A substance has a half-life of 4 days. What fraction of the original amount remains after 4 days?

a. $\frac{5}{8}$

b. 1

c. $\frac{1}{4}$

d. $\frac{1}{2}$ ✓

Since $t/4 = 1$ half-lives have passed: fraction remaining $= \left(\frac{1}{2}\right)^1 = \frac{1}{2}$.

41 A substance has a half-life of 2 days. What fraction of the original amount remains after 10 days?

a. $\frac{5}{32}$

b. $\frac{1}{64}$

c. $\frac{1}{16}$

d. $\frac{1}{32}$ ✓

Since $t/2 = 5$ half-lives have passed: fraction remaining $= \left(\frac{1}{2}\right)^5 = \frac{1}{32}$.

42 A substance has a half-life of 5 days. What fraction of the original amount remains after 25 days?

a. $\frac{5}{32}$

b. $\frac{1}{64}$

c. $\frac{1}{16}$

d. $\frac{1}{32}$ ✓

Since $t/5 = 5$ half-lives have passed: fraction remaining $= \left(\frac{1}{2}\right)^5 = \frac{1}{32}$.

43 A substance has a half-life of 4 days. What fraction of the original amount remains after 12 days?

a. $\frac{1}{8}$ ✓

b. $\frac{1}{16}$

c. $\frac{1}{4}$

d. $\frac{3}{8}$

Since $t/4 = 3$ half-lives have passed: fraction remaining $= \left(\frac{1}{2}\right)^3 = \frac{1}{8}$.

44 A substance has a half-life of 9 days. What fraction of the original amount remains after 45 days?

- a. $\frac{1}{64}$
- b. $\frac{1}{16}$
- c. $\frac{1}{32}$ ✓
- d. $\frac{5}{32}$

Since $t/9 = 5$ half-lives have passed: fraction remaining $= \left(\frac{1}{2}\right)^5 = \frac{1}{32}$.

Multiple choice — hard 15

45 An investment grows from 300 dollars to 507 dollars in 2 years under $A(t) = A_0(1 + r)^t$. Find the annual growth rate r .

- a. $\frac{3}{10}$ ✓
- b. $\frac{2}{5}$
- c. $-\frac{3}{10}$
- d. $\frac{13}{10}$

$507 = 300(1 + r)^2$: $(1 + r)^2 = \frac{507}{300}$, so $1 + r = \frac{13}{10}$, giving $r = \frac{3}{10}$.

46 An investment grows from 200 dollars to 128 dollars in 2 years under $A(t) = A_0(1 + r)^t$. Find the annual growth rate r .

- a. $-\frac{1}{10}$
- b. $-\frac{1}{5}$ ✓
- c. $\frac{1}{5}$
- d. $\frac{4}{5}$

$128 = 200(1 + r)^2$: $(1 + r)^2 = \frac{128}{200}$, so $1 + r = \frac{4}{5}$, giving $r = -\frac{1}{5}$.

- 47 An investment grows from 100 dollars to 169 dollars in 2 years under $A(t) = A_0(1 + r)^t$. Find the annual growth rate r .

a. $-\frac{3}{10}$

b. $\frac{3}{10}$ ✓

c. $\frac{2}{5}$

d. $\frac{13}{10}$

$$169 = 100(1 + r)^2: (1 + r)^2 = \frac{169}{100}, \text{ so } 1 + r = \frac{13}{10}, \text{ giving } r = \frac{3}{10}.$$

- 48 An investment grows from 200 dollars to 98 dollars in 2 years under $A(t) = A_0(1 + r)^t$. Find the annual growth rate r .

a. $-\frac{3}{10}$ ✓

b. $\frac{3}{10}$

c. $\frac{7}{10}$

d. $-\frac{1}{5}$

$$98 = 200(1 + r)^2: (1 + r)^2 = \frac{98}{200}, \text{ so } 1 + r = \frac{7}{10}, \text{ giving } r = -\frac{3}{10}.$$

- 49 An investment grows from 200 dollars to 162 dollars in 2 years under $A(t) = A_0(1 + r)^t$. Find the annual growth rate r .

a. $\frac{9}{10}$

b. $\frac{1}{10}$

c. 0

d. $-\frac{1}{10}$ ✓

$$162 = 200(1 + r)^2: (1 + r)^2 = \frac{162}{200}, \text{ so } 1 + r = \frac{9}{10}, \text{ giving } r = -\frac{1}{10}.$$

50 Find the horizontal asymptote and y-intercept of $h(x) = -7(2)^x - 3$.

- a. Horizontal asymptote $y = -10$; y-intercept (-3) .
- b. Horizontal asymptote $y = -3$; y-intercept (-7) .
- c. **Horizontal asymptote $y = -3$; y-intercept (-10) . ✓**
- d. Horizontal asymptote $y = 0$; y-intercept (-10) .

As $x \rightarrow \pm \infty$ (in the direction $2^x \rightarrow 0$), $h(x) \rightarrow -3$, giving horizontal asymptote $y = -3$. y-intercept: $h(0) = -7(1) - 3 = -10$, so $(0, -10)$.

51 Find the horizontal asymptote and y-intercept of $h(x) = 5(2)^x + 1$.

- a. Horizontal asymptote $y = 6$; y-intercept (1) .
- b. **Horizontal asymptote $y = 1$; y-intercept (6) . ✓**
- c. Horizontal asymptote $y = 1$; y-intercept (5) .
- d. Horizontal asymptote $y = 0$; y-intercept (6) .

As $x \rightarrow \pm \infty$ (in the direction $2^x \rightarrow 0$), $h(x) \rightarrow 1$, giving horizontal asymptote $y = 1$. y-intercept: $h(0) = 5(1) + 1 = 6$, so $(0, 6)$.

52 Find the horizontal asymptote and y-intercept of $h(x) = -3(2)^x - 7$.

- a. Horizontal asymptote $y = -7$; y-intercept (-3) .
- b. **Horizontal asymptote $y = -7$; y-intercept (-10) . ✓**
- c. Horizontal asymptote $y = 0$; y-intercept (-10) .
- d. Horizontal asymptote $y = -10$; y-intercept (-7) .

As $x \rightarrow \pm \infty$ (in the direction $2^x \rightarrow 0$), $h(x) \rightarrow -7$, giving horizontal asymptote $y = -7$. y-intercept: $h(0) = -3(1) - 7 = -10$, so $(0, -10)$.

53 Find the horizontal asymptote and y-intercept of $h(x) = 3(2)^x - 4$.

a. Horizontal asymptote $y = -1$; y-intercept (-4) .

b. Horizontal asymptote $y = -4$; y-intercept (-1) . ✓

c. Horizontal asymptote $y = -4$; y-intercept (3) .

d. Horizontal asymptote $y = 0$; y-intercept (-1) .

As $x \rightarrow \pm \infty$ (in the direction $2^x \rightarrow 0$), $h(x) \rightarrow -4$, giving horizontal asymptote $y = -4$. y-intercept: $h(0) = 3(1) - 4 = -1$, so $(0, -1)$.

54 Find the horizontal asymptote and y-intercept of $h(x) = 7(2)^x + 5$.

a. Horizontal asymptote $y = 12$; y-intercept (5) .

b. Horizontal asymptote $y = 5$; y-intercept (12) . ✓

c. Horizontal asymptote $y = 0$; y-intercept (12) .

d. Horizontal asymptote $y = 5$; y-intercept (7) .

As $x \rightarrow \pm \infty$ (in the direction $2^x \rightarrow 0$), $h(x) \rightarrow 5$, giving horizontal asymptote $y = 5$. y-intercept: $h(0) = 7(1) + 5 = 12$, so $(0, 12)$.

55 A savings account earns interest modeled by $A(t) = A_0(1 + \frac{1}{20})^{nt}$, where $nt = 15$. If 2000 dollars is deposited, find the balance to the nearest dollar.

a. 4008

b. 4258

c. 2000

d. 4158 ✓

$A = 2000(1 + \frac{1}{20})^{15} \approx 2000(2.0789) \approx 4158$ dollars.

- 56 A savings account earns interest modeled by $A(t) = A_0(1 + \frac{1}{25})^{nt}$, where $nt = 9$. If 5000 dollars is deposited, find the balance to the nearest dollar.

a. 7167

b. 7117 ✓

c. 5000

d. 7067

$$A = 5000(1 + \frac{1}{25})^9 \approx 5000(1.4233) \approx 7117 \text{ dollars.}$$

- 57 A savings account earns interest modeled by $A(t) = A_0(1 + \frac{3}{100})^{nt}$, where $nt = 6$. If 5000 dollars is deposited, find the balance to the nearest dollar.

a. 5970 ✓

b. 6120

c. 5920

d. 5000

$$A = 5000(1 + \frac{3}{100})^6 \approx 5000(1.1941) \approx 5970 \text{ dollars.}$$

- 58 A savings account earns interest modeled by $A(t) = A_0(1 + \frac{1}{25})^{nt}$, where $nt = 13$. If 1000 dollars is deposited, find the balance to the nearest dollar.

a. 1000

b. 1515

c. 1665 ✓

d. 1765

$$A = 1000(1 + \frac{1}{25})^{13} \approx 1000(1.6651) \approx 1665 \text{ dollars.}$$

59 A savings account earns interest modeled by $A(t) = A_0(1 + \frac{1}{100})^{nt}$, where $nt = 16$. If 4000 dollars is deposited, find the balance to the nearest dollar.

a. 4000

b. 4590

c. 4740

d. 4690 ✓

$$A = 4000(1 + \frac{1}{100})^{16} \approx 4000(1.1726) \approx 4690 \text{ dollars.}$$